Brushless Lab

Reverse-engineer and service brushless motors in VR



A six-bench virtual workshop: take a motor apart, measure it, find the fault, repair or rewind it, and prove it on a test stand.

Godot 4.7 · Meta Quest and desktop · October 2026

THE PROBLEM

Everywhere, yet hard to learn

Everywhere

Drones, e-bikes, power tools, fans, pumps, HVAC, robots, hard drives, electric cars.

Hard to learn hands-on

Practice motors cost money, get damaged, and a real fault is rarely there on the day you need it.

Invisible physics

Flux, back-EMF, current and bearing defects can't be seen. Students learn formulas without ever seeing what they describe.

Skills that matter

Reverse-engineering an unknown motor, diagnosing it and rewinding it are real workshop skills, and they are rarely taught.

THE IDEA

One motor, six benches



1	2	3	4	5	6
Job Desk	**Teardown**	**Electrical**	**Inspect & Repair**	**Winding**	**Test Stand**
ticket, datasheet, report	right tool, right order	R, L, insulation, back-EMF	magnets, coils, shaft	rewind, star of slots	ESC, spectra, final test

- Everything you measure goes into a **notebook**, and the desk grades your report against the hidden truth.
- Mistakes have consequences: the wrong tool, a bearing pressed on the wrong ring, a forgotten shim or a magnet glued the wrong way round are all recorded.
- **Guided** mode shows the next step and explains it; **Expert** mode shows nothing.

WHAT YOU DO

Three kinds of job

Reverse-engineer

A mystery outrunner with its label worn off. Work out the teeth, magnets, winding layout, star or delta, KV, resistance and stator size, so a winder could copy it.

10 real slot/pole combinations, generated from a seed

Service

A customer's motor and a complaint ("grinding noise", "runs hot", "trips the controller"). Find the hidden fault, repair it, and pass the final test.

9 faults, or none: "customer says it's weak"

Free lab

A healthy motor to explore: every instrument, every part, the test stand.

for a first visit or for teaching

Motors on the bench

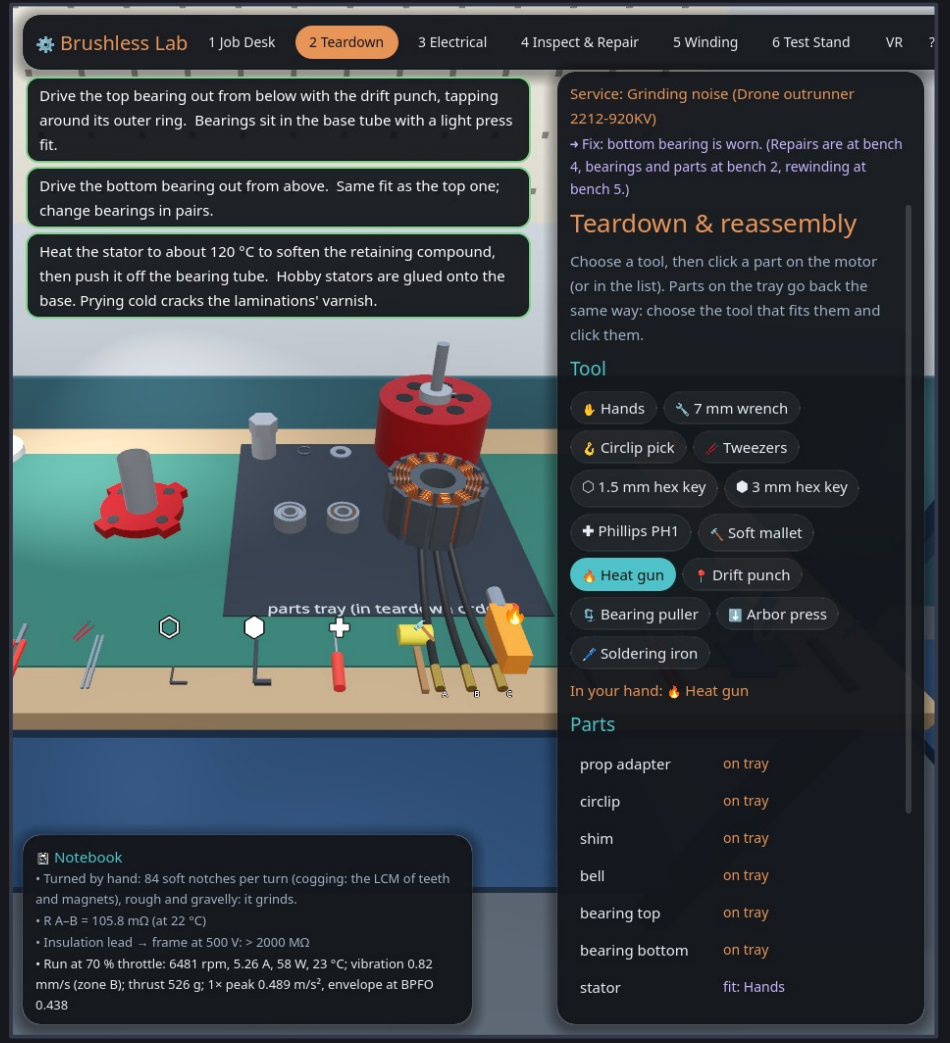
- **Drone outrunner 2212-920KV**: 12 teeth, 14 magnets, star, sensorless, 10×4.5 propeller on 3S.
- **Industrial inrunner 57BL-24V**: 190 KV, Hall sensors, 608 bearings, dynamometer brake.

BENCH 2

Teardown and reassembly

- Choose a tool, then click a part. It only comes off if nothing still holds it and the tool fits.
- Each step explains the shop-floor reason, e.g. heat the stator to 120 °C before pushing it off, because prying it cold cracks the varnish.
- Hand checks: turn the shaft (84 cogging notches on 12N14P, grinding, scraping), spin and rock each bearing.
- Parts go onto a tray in order and go back on in reverse.

Thirteen tools, two motor designs, and the order is enforced, as it would be on a real bench.

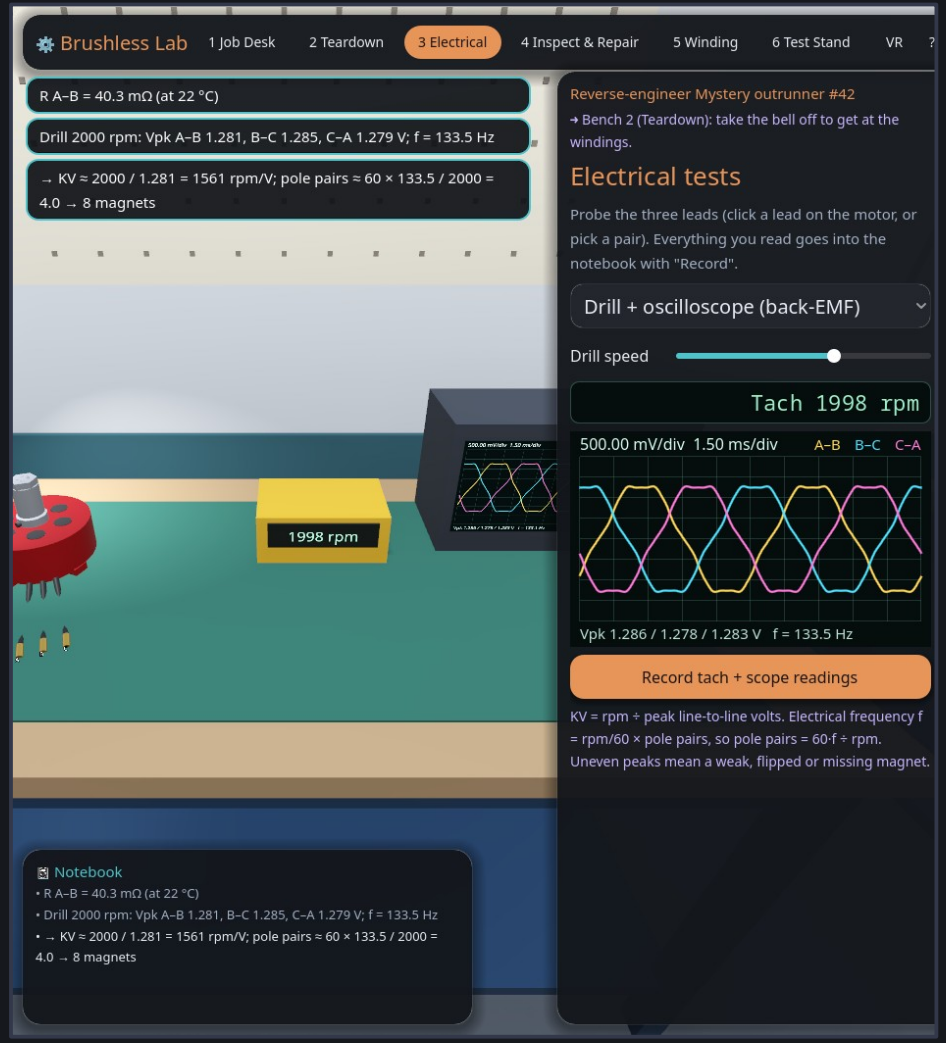


BENCH 3

Electrical tests

- 4-wire milliohm meter vs a normal multimeter: shows why 0.1 Ω windings need the right instrument.
- LCR meter at 1 kHz, 500 V insulation tester to the frame, Hall sensor tester.
- Spin it with a drill and read the scope: KV = rpm ÷ peak line volts; pole pairs = 60·f ÷ rpm.
- Uneven EMF peaks reveal a weak, flipped or missing magnet.

Readings come from the motor's real condition: a shorted phase makes two lead pairs read low.

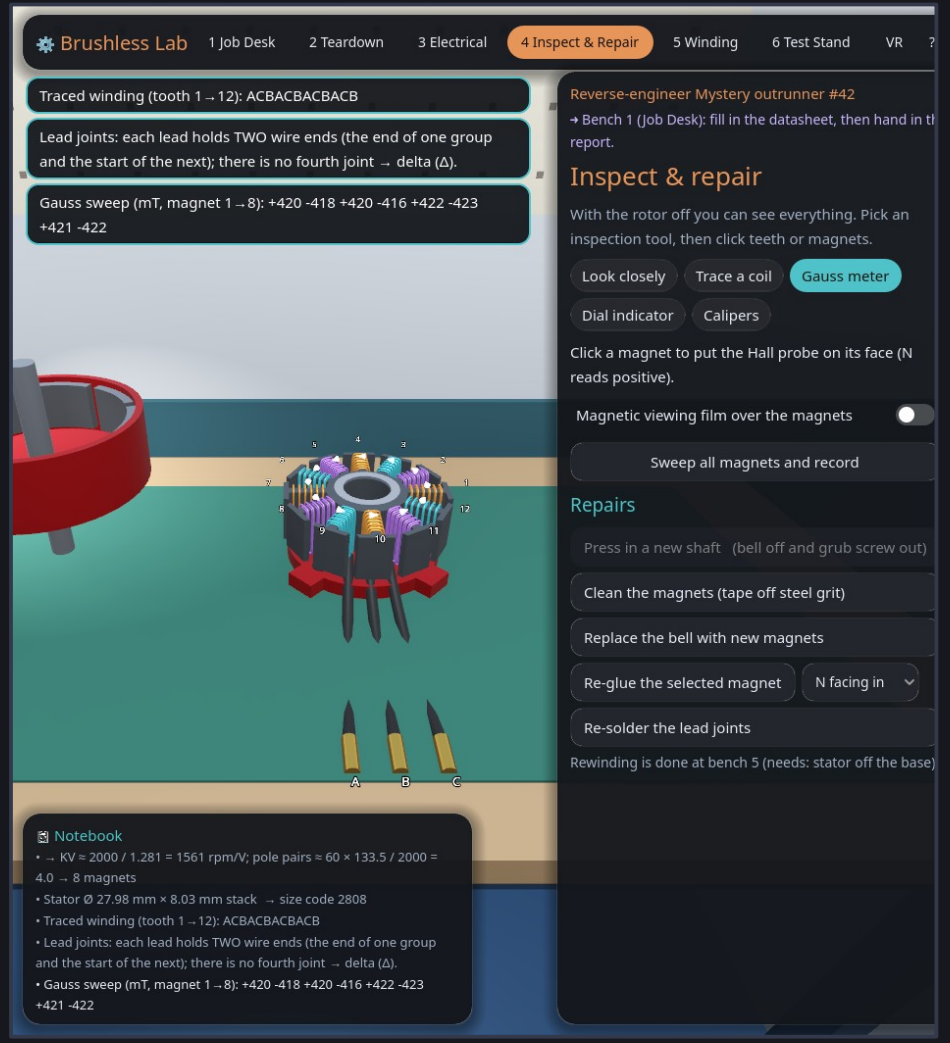


BENCH 4

Inspect and repair

- Count teeth and magnets, measure the stator with calipers (size code, e.g. 2808).
- Trace each coil's phase and direction, read the lead joints: star or delta.
- Gauss meter or magnetic viewing film on every magnet; dial indicator for shaft runout.
- Repairs: bearings, shaft, re-glue a magnet (the right way round), new bell, clean grit, re-solder joints, Hall sensor.

Inspection confirms what the instruments suggested; nothing is labelled for you.



BENCH 5

Rewinding

- Wind each tooth with a phase letter: A B C clockwise, a b c counter-clockwise.
- The star of slots shows every coil's EMF phasor and the three phase sums: they must be equal and 120° apart.
- Winding factor computed live (0.933 for the classic 12N14P AabBCcaABbcC).
- Turns per tooth and star/delta set the KV: copy the original or the motor comes back different.

A wrong coil is caught by the physics: unbalanced phases, a lower KV, a failed test.

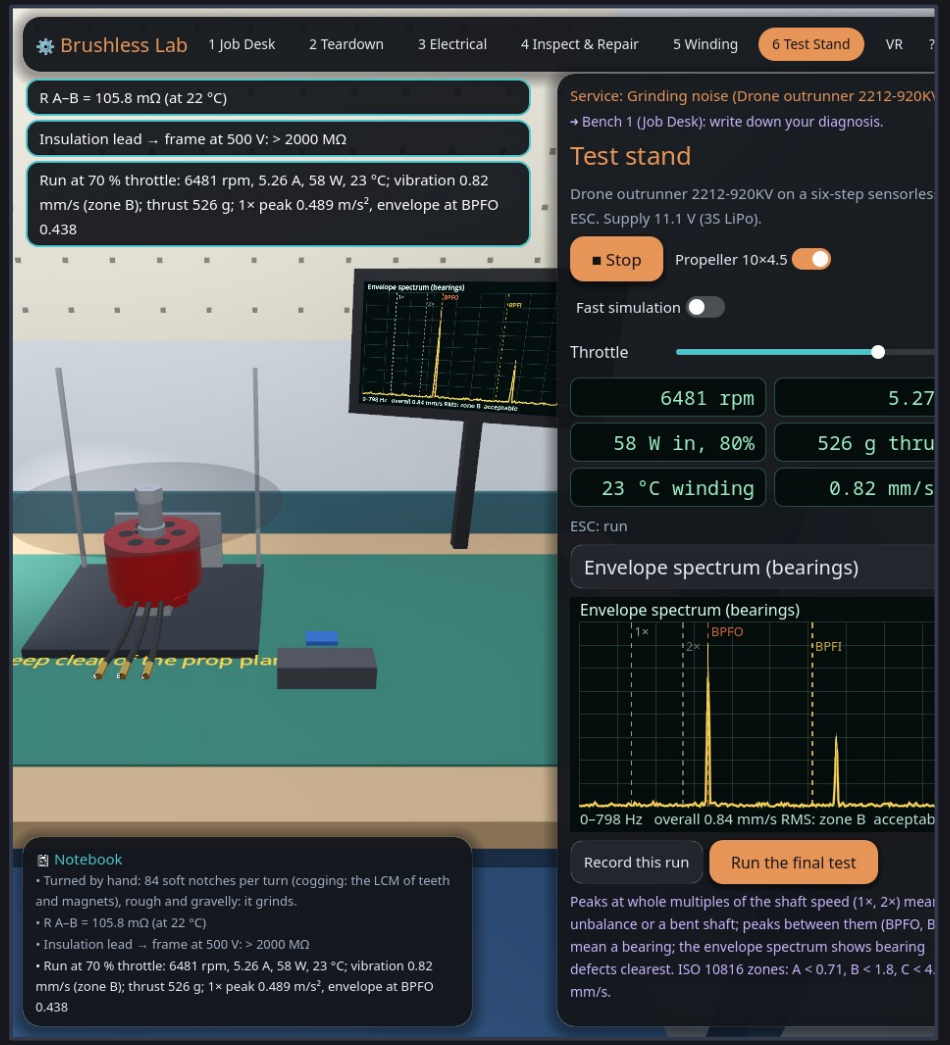


BENCH 6

Test stand

- Six-step ESC: sensorless start-up and back-EMF zero crossings, or Hall sensors.
- Live speed, current, power, efficiency, thrust or torque, winding temperature, scope, and the motor's sound.
- Vibration spectrum with shaft (1×, 2×) and bearing (BPFO, BPFI) markers, the envelope spectrum, and the ISO 10816 zone.
- Final test: starts, no-load speed ≈ KV × V, low current, zone A/B, cool, balanced, insulated.

A worn bearing shows up as a line at its defect frequency, not as a scripted message.



SERVICE JOBS

Nine faults, found the way a technician finds them

Worn bearing

grinds by hand; BPFO line in the envelope spectrum

Bent shaft

big 1× vibration peak (zone D); dial indicator runout

Shorted turns

two lead pairs read low, inductance lower still; runs hot

Open phase

two lead pairs read open; stutters, loses sync

Insulation fault

megger well below 1 MΩ; the controller trips

Loose magnet

rattle and knock once per turn; Gauss sweep shifted

Demagnetised

KV too high, weak torque; low Gauss readings

Metal debris

scraping by hand; grit visible on the magnets

Dead Hall sensor

jerky start; the tester shows a stuck code (57BL)

Symptoms are not scripted: each fault changes the motor's state, and every instrument reads from it.

WHERE THE NUMBERS COME FROM

Computed, not scripted

Design
slots, poles, turns, wire, magnets, bearings

Condition
wear, damage, repairs, your rewind

Flux model
magnet arcs → tooth flux → λ(θ) → back-EMF and torque

Drive simulation
six-step ESC; circuit and rotor solved every 20 µs

Instruments
meters, scope, spectra, sound, hand feel

- **Standard methods**: star-of-slots windings, back-EMF from flux linkage, bearing defect frequencies from the real ball geometry, ISO 10816 severity zones.
- **Calibrated** once: the flux scale is set so a healthy motor's back-EMF matches its rated KV. Everything else follows.
- **Typical values, not measurements**: friction, inertia, heat flow and vibration amplitudes are realistic for these sizes but hand-tuned.
- 70 physics checks: KV 920 from the EMF, 84 cogging steps, no-load ≈ KV × V, 550–1000 g thrust, each fault's signature.

PERFORMANCE

Physics that fits on a headset

Detailed model

20 µs steps, back-EMF zero-crossing detector modelled

≈ 125 ms

desktop CPU per simulated second

Fast model (Quest)

100 µs steps, ESC assumed in sync with the rotor

≈ 45 ms

about 3× cheaper

Same answers

speed within 3–4 %, current within 8 %

same fault signatures: repaired motors pass and unrepaired ones fail on both

- The simulation runs within a time budget each frame. If a computer can't keep up, it runs in slow motion and says so.
- A desktop that falls behind switches to the fast model by itself; there is also a switch on the test stand.
- Every 3D part is built in code from the motor's design, so there are no model files to load and the app stays small (Quest APK ≈ 90 MB).

FEEDBACK

How it is graded

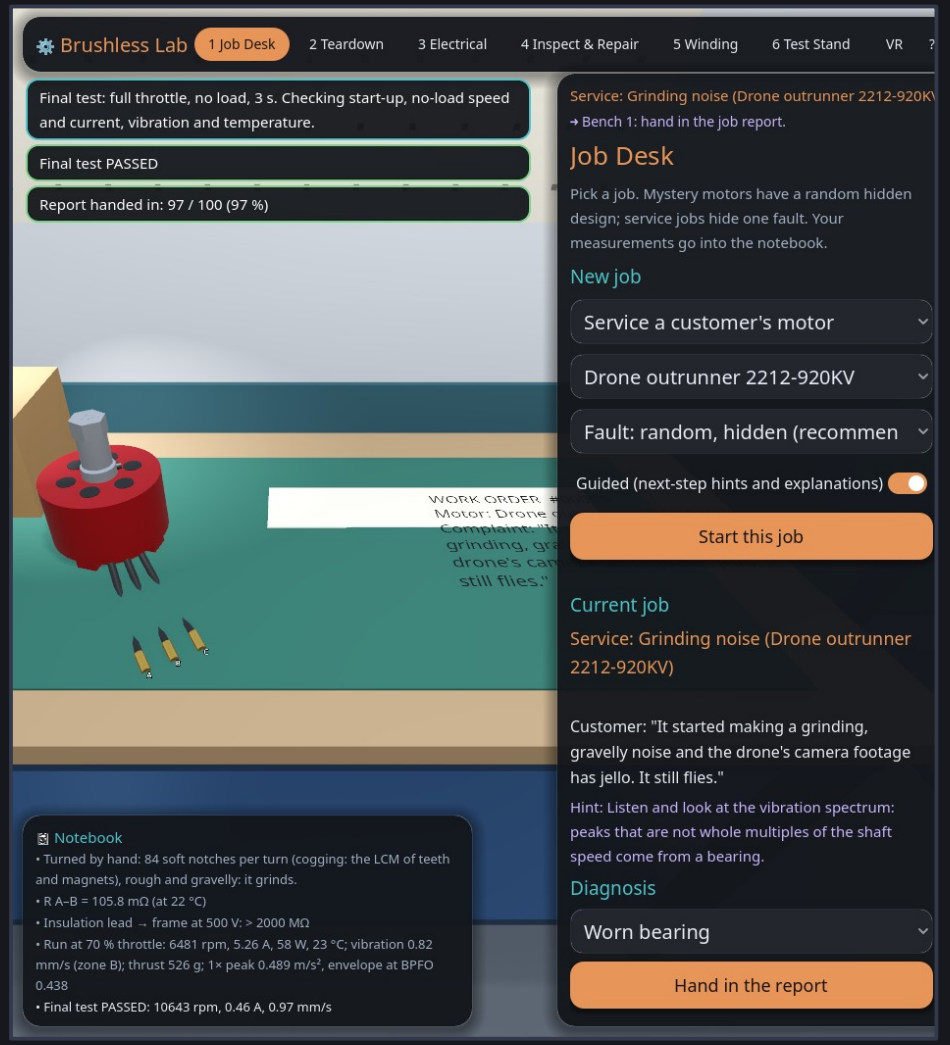
40
diagnosis

40
repair: nothing left wrong

20
final test passed

−3
per workshop mistake (max −20)

- Reverse-engineering: each datasheet field (teeth, magnets, winding, connection, KV, resistance, size) against the real design.
- The report lists what was right, what was wrong and the true answer.



STATUS

Built and tested, and what is still missing

Done

- Six benches, three motors (one generated), nine faults, grading.
- Desktop executable (Linux) and a Quest build.
- 70 automatic physics checks and 60 end-to-end checks that play complete jobs through the real interface and take screenshots.
- README with the method and the source of every number.

Not yet done

- Tried on a real headset: frame rate, controller feel and the VR panels need a Quest session.
- Checked against a real motor: measure a 2212 (KV, resistance, thrust, vibration) and tune the constants.
- Field-oriented control (FOC) drives; jobs with two faults at once.
- Tried with students: usability and learning gains.

A training simulator, not a design tool: absolute numbers are realistic in size, not exact.

PLAN

Next steps

Step 1

Headset

Install on Quest 3S; frame rate on the test stand; VR panel comfort.

Step 2

Real bench data

Measure a real 2212 and a 57BL; fit friction, heat and vibration constants.

Step 3

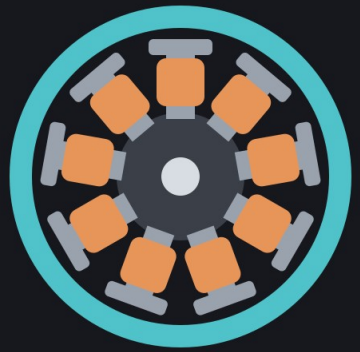
Class trial

A lab session with students: pre/post quiz, time per job, feedback.

Step 4

More content

FOC drive, two-fault jobs, more motor sizes, an instructor view.



Brushless Lab

Open the motor. Measure it. Fix it. Prove it.

Thank you. Questions are welcome.